SE-LAB 7 FUZZ TESTING

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SEC-F

1. Execution of Original Application

```
PS C:\Users\Admin\OneDrive\Desktop\SE-LAB-PES1UG24AM340\SE-LAB7\ride-share-lab7\Lab 7> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan:12
Cab and Distance: ('Sedan', 12.0)

==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): 120,95,210
Average Fare: 141

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): 250|30
Payable Fare: 265.0
```

2. Initial Fuzz Testing Results

```
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
=========================== short test summary info ===========================
FAILED test_ride_share_fare.py::test_parse_trip_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_ride_share_fare.py::test_fare_average_no_crash - BaseExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_ride_share_fare.py::test_payable_fare_no_crash - BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed, 14 warnings in 19.82s ==============================
```

3. Execution of Fixed Application

```
PS C:\Users\Admin\OneDrive\Desktop\SE-LAB-PES1UG24AM340\SE-LAB7\ride-share-lab7\Lab 7> python -m pytest test_ride_share_fare.py -v
============================= test session starts =============================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Admin\OneDrive\Desktop\SE-LAB-PES1UG24AM340\SE-LAB7\ride-share-lab7\Lab 7
plugins: anyio-4.14.2, hypothesis-6.170.15, langsmith-0.14.4, platformdirs-4.12.4, typeguard-4.6.0
collected 3 items

test_ride_share_fare.py::test_parse_trip_no_crash PASSED                 [ 33%]
test_ride_share_fare.py::test_fare_average_no_crash PASSED               [ 66%]
test_ride_share_fare.py::test_payable_fare_no_crash PASSED               [100%]

============================== warnings summary ===============================
..\..\..\..\..\..\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Admin\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 h
as been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

..\..\..\..\..\..\AppData\Roaming\Python\Python314\site-packages\langsmith\_internal\_beta_decorator.py:70: 13 warnings
  C:\Users\Admin\AppData\Roaming\Python\Python314\site-packages\langsmith\_internal\_beta_decorator.py:70: DeprecationWarning: 'asyncio.iscorou
tinefunction' is deprecated and slated for removal in Python 3.16; use inspect.iscoroutinefunction() instead
    if asyncio.iscoroutinefunction(func):

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 3 passed, 14 warnings in 3.39s =======================
```

4. Final Fuzz Testing Results

```
PS C:\Users\Admin\OneDrive\Desktop\SE-LAB-PES1UG24AM340\SE-LAB7\ride-share-lab7\Lab 7> python ride_share_fare.py
==== Running parse_trip ====
Enter cab:distance in km (e.g. sedan:12): sedan:12
Cab and Distance: ('Sedan', 12.0)

==== Running fare_average ====
Enter fares as CSV (e.g. 120,95,210): 120,95,210
Average Fare: 142

==== Running payable_fare ====
Enter fare|promo (e.g. 250|30): 250|30
Payable Fare: 265.0
```